BLOUK EDI Order File Generator

A browser-based tool for Bloomsbury Publishing that converts subscription order data from Excel into the fixed-width EDI format required by the TJClays POD Order Importer and separately generates XML metadata files from a streamlined three-column spreadsheet.

No installation, server, or build tools required — the entire application runs in the browser from a single folder.

Table of Contents

- BLOUK EDI Order File Generator
  - Table of Contents
  - Overview
  - File Structure
  - Quick Start
  - EDI Order File Generator
    - How to Use
    - Order Template — Column Reference
    - Order Template — Structure
    - Order Template — Text Formatting
    - Country Code Validation
    - Row Grouping
    - File Settings Reference
    - Order Number Scheme
    - Column Mapping Panel
    - EDI Output Format
    - Record Type Reference
      - $$HDR — File Header
      - H1 — Order Header (350 characters)
      - H2 — Customer / Address (358 characters)
      - H3 — Payment Terms (20 characters)
      - D1 — Line Item (266 characters)
      - $$EOF — File Footer
    - Output Filename Convention
  - XML Metadata Generator
    - How to Use
    - Metadata Template — Column Reference
    - ISSN Validation
    - Derived Values
    - Spine Size Calculation
    - XML Output Structure
    - Summary Report
  - Column Validation
  - Template Downloads

- Dependencies
- Browser Compatibility
- Development Notes
  - EDI Field Positions
  - Adding a New EDI Column
  - Changing Fixed XML Metadata Values
  - Changing the Paper / Extent Threshold
  - Changing the Spine Formula
  - Changing the Carrier Code
  - Updating the Supported Country Code List

Overview

The application provides two independent tools in a single interface:

EDI Order File Generator — accepts a 16-column order spreadsheet and produces a plain-text fixed-width EDI file in the BLOUK T1 format.

XML Metadata Generator — accepts a simplified 3-column spreadsheet (ISSN, Title, Page Extent) and produces one XML metadata file per row. All remaining specification values — trim size, binding, lamination, paper type, and spine width — are derived automatically from the page extent using defined business rules. The XML files are bundled into `metadata.zip`, and a plain-text summary report (`metadata_summary.txt`) is downloaded alongside it.

Both tools enforce strict column validation on upload: if the uploaded file does not exactly match the expected template structure, the file is rejected before any data is processed and a detailed per-column error report is shown.

File Structure

```
blouk_edi/
├── index.html              — Application markup
├── style.css               — All visual styling and layout
├── script.js               — All application logic
├── order_file.xlsx         — Order upload template (download from app)
├── metadata_template.xlsx  — Metadata upload template (download from
app)
├── USERGUIDE.pdf           — End-user guide (download from app)
└── README.md               — This document
```

All logic is in `script.js`; all presentation is in `style.css`; `index.html` contains only semantic structure. The two `.xlsx` template files and `USERGUIDE.pdf` must remain in the same root directory as `index.html` for the in-app download buttons to work.

Quick Start

1. Copy the entire `blouk_edi/` folder to any location on your machine or a web server.
2. Open `index.html` in a modern web browser (Chrome, Edge or Firefox recommended).
3. An internet connection is required on first load to fetch CDN libraries. After that, the browser cache will serve them offline.

EDI Order File Generator

How to Use

Step 1 — Download the template (first time only) Click **Download Order Template** in the Import Excel Data panel to save `order_file.xlsx`. A **Download User Guide** button is also available directly below it. Use the template as the basis for every upload — do not rename, reorder or add columns.

Step 2 — Populate the template Fill in your subscription order rows starting from row 4 (the first three rows are the title bar, column group labels, and column headers respectively). Row 4 contains a pre-filled example row in grey italics — overwrite it or start from row 5. See Order Template — Column Reference for field details.

Step 3 — Upload the file Drag `order_file.xlsx` onto the upload zone or click to browse. The app validates that the column structure exactly matches the template. If validation passes, a green confirmation shows the row count and the Source Data tab populates with a preview table.

Step 4 — Adjust File Settings (if required) Expand the **File Settings** panel to review or update the batch configuration. The Order Number Start field is read-only and auto-populated — no action is needed.

Step 5 — Review Column Mapping (optional) The **Column Mapping** panel shows how spreadsheet columns are mapped to EDI fields. Auto-mapping occurs on upload; expand the panel to override any mapping manually.

Step 6 — Generate Click **Generate EDI File**. The app validates all country codes first — if any row contains an unrecognised value, generation is aborted and a red error panel lists every affected row with its Order Ref and the invalid value. If validation passes, the app groups rows into orders, builds all EDI records, switches to the EDI Preview tab with colour-coded output, and enables the Download button.

Step 7 — Download Click **Download .txt** to save the EDI file using the standard filename convention.

Order Template — Column Reference

The order template contains 16 columns in a fixed order. All column names must be preserved exactly, including any trailing spaces.

#	Column Header	EDI Field	Notes
1	`Order Ref`	Order / item reference	Groups rows into a single order when shared with the same delivery name and company. Leading zeros preserved.

#	Column Header	EDI Field	Notes
2	`ISSN (13-digit)`	ISBN / ISSN (D1)	Stored and passed through as a 13-digit text string. Standard ISSNs are 8 digits — pad to 13 as required by your workflow.
3	`Journal/ Issue Title`	Title	Not written to EDI; for reference only (note: two spaces after `/`)
4	`Volume Number`	—	Not written to EDI; for reference only (note: trailing space)
5	`Volume Part`	—	Not written to EDI; for reference only
6	`Year`	—	Not written to EDI; for reference only
7	`Quantity`	D1 quantity	Falls back to Default Qty setting if blank or zero
8	`Delivery Name`	H2 contact name	Note: trailing space
9	`Delivery Company name`	H2 city / company	
10	`Delivery address line 1`	H2 address line 1	
11	`Delivery address line 2`	H2 address line 2	
12	`Delivery address line 3`	H2 address line 3	
13	`Delivery Country`	H2 country code	Must be a recognised ISO 2-letter or alpha-3 code; validated before generation. See Country Code Validation.
14	`Post code`	H2 post code	
15	`Telephone number`	H2 telephone	Leading zeros preserved. Note: trailing space.
16	`Email address`	H2 email	

> **Important:** Several column headers contain trailing spaces or non-obvious punctuation (such as the double space in `Journal/ Issue Title`). These are significant — they are part of the exact

> header string validated on upload. Always use `order_file.xlsx` as your starting point and do not modify the header row.

Order Template — Structure

The current `order_file.xlsx` template has three rows above the data entry area:

Row	Content
1	Title banner (`BLOUK EDI Order Form — Bloomsbury Publishing`)
2	Column group labels (ORDER, PUBLICATION, DELIVERY, etc.)
3	Column headers — the field names validated on upload
4+	Data rows (row 4 contains a pre-filled example in grey italics)

The application reads headers from **row 3** and data from **row 4 onwards**. Rows 1 and 2 are skipped automatically. The template also includes a **Country Codes** sheet listing all supported ISO codes for reference.

Order Template — Text Formatting

All data columns in the template are pre-formatted as **Text** (`@` format) in Excel. This ensures:

- Leading zeros in Order Ref are preserved exactly as entered (e.g. `0027816` is not converted to `27816`)
- The full 13-digit ISSN is stored without numeric rounding or truncation
- Telephone numbers with leading zeros are preserved (e.g. `07700900000`)
- Postcodes containing letters or spaces are not altered

Do not change the cell format of any column. If formatting is lost, download a fresh copy of the template from the app.

Country Code Validation

The **Delivery Country** column is validated at two points:

In the spreadsheet — column M has a dropdown data validation listing all supported codes. Excel's `Stop` error style prevents entry of unrecognised values. A tooltip appears when the cell is selected explaining the expected format.

At generation time — before any EDI output is written, `generateEDI()` checks every order's country code against the `VALID_ALPHA3` set (derived from `ISO_2_TO_3` in `script.js`). If any code fails, generation is aborted entirely and a structured error panel is shown listing each affected row with its row number, invalid value, and Order Ref.

Accepted formats:

Format	Example	Notes
ISO 3166-1 alpha-2 (2-letter)	`GB`	Automatically converted to alpha-3 for the H2 record
ISO 3166-1 alpha-3 (3-letter)	`GBR`	Must be a code present in `VALID_ALPHA3` — not any 3-letter string

City or region codes such as `LON`, `NYC`, or `EUR` are explicitly rejected at both validation points.

The supported country list covers 46 territories. See `ISO_2_TO_3` in `script.js` for the complete mapping, or the **Country Codes** sheet in `order_file.xlsx` for a human-readable reference.

Row Grouping

Rows that share the same **Order Ref**, **Delivery Company name** and **Delivery Name** are automatically grouped into a single order with multiple D1 line items. Each unique combination produces its own H1 / H2 / H3 block with sequential D1 records beneath it.

Rows with no Order Ref value are each treated as a separate single-line order.

File Settings Reference

The **File Settings** panel (collapsed by default) contains EDI batch configuration. Expand it by clicking the panel header.

Setting	Default	Editable	Description
File ID / Batch Number	`0027816`	Yes	Written into `$$HDR` and `$$EOF` markers. Update per batch if required.
File Prefix	`PO`	No	Used in the output filename only (e.g. `PO.0027816_...`).
Sender Code	`BLOO`	No	4-character code written to `$$HDR` and `$$EOF`.
Currency	`GBP`	Yes	3-character ISO currency code written to each H1 record.
Payment Terms	`FCA`	Yes	Incoterms code written to each H3 record. `FCA` = Free Carrier; `DAP` = Delivered at Place.
Order Number Start	Auto	No	Timestamp-derived 10-digit seed. Regenerated on each Generate click.
Default Qty	`1`	Yes	Fallback quantity when the Quantity column is blank or zero.

Order Number Scheme

Order numbers are derived from the date and time at the moment **Generate EDI File** is clicked, ensuring uniqueness across batches without any manual coordination.

Format: `3` + `YY` + `MM` + `DD` + `HH` + `MM` (10 digits)

Example: Clicking Generate at 14:23 on 24 February 2026 produces seed `3260224142`.

Orders within the same batch increment sequentially from that seed: `3260224142`, `3260224143`, `3260224144`, and so on.

Provided two batches are not generated within the same calendar minute, their order number ranges will never overlap.

Column Mapping Panel

The **Column Mapping** panel (collapsed by default) shows the relationship between each EDI field and the corresponding spreadsheet column. Auto-mapping occurs on upload using exact header matching.

To override any mapping, expand the panel and use the dropdowns. Selecting `(none)` leaves that EDI field blank (padded with spaces or zeros) in the output.

Changes take effect on the next Generate click.

EDI Output Format

The generated file is a fixed-width ASCII plain-text file with CRLF (`\r\n`) line endings. Fields occupy exact character positions with no delimiters. Text fields are space-padded on the right; numeric fields are zero-padded on the left.

File structure:

```
$$HDR{sender}  {fileId}   {timestamp}
H1...  (350 chars)
H2...  (358 chars)
H3...  ( 20 chars)
D1...  (266 chars each — one per line item)
[H1 / H2 / H3 / D1 blocks repeat for each order]
$$EOF{sender}  {fileId}   {timestamp}{recordCount}
```

The `$$EOF` record count is a 7-digit zero-padded total of H1 + H2 + H3 + D1 lines only. `$$HDR` and `$$EOF` themselves are not included in the count.

Record Type Reference

$$HDR — File Header

Chars	Content
0–4	`$$HDR`
5–8	Sender code (`BL00`)
9–10	Two spaces

Chars	Content
11–17	File ID (7 chars, zero-padded)
18–20	Three spaces
21–34	Timestamp `YYYYMMDDHHMMSS`

H1 — Order Header (350 characters)

Position	Length	Content
0	2	`H1`
2	15	Order number
17	8	Order date `YYYYMMDD`
25	30	Spaces
55	1	`C` (currency flag)
56	1	Space
57	28	Customer / order reference
85	7	Spaces
92	8	Carrier code — `RMA` (Royal Mail)
100	2	`N`
102	30	Spaces
132	57	Zeros
189	6	Spaces
195	40	PDF placeholder (`.PDF` padded to 40 chars)
235	2	Spaces
237	3	Currency code (e.g. `GBP`)
240	110	Spaces

H2 — Customer / Address (358 characters)

Position	Length	Content
0	2	`H2`
2	15	Order number
17	27	Customer code (`ST` + subscription ref)
44	50	Contact name

Position	Length	Content
94	50	Address line 1
144	50	Address line 2
194	50	Address line 3
244	50	Email address
294	32	City / company name
326	9	Post code
335	3	Country code (ISO alpha-3)
338	20	Telephone number

H3 — Payment Terms (20 characters)

Position	Length	Content
0	2	`H3`
2	15	Order number
17	3	Incoterms code (`FCA` or `DAP`)

D1 — Line Item (266 characters)

Position	Length	Content
0	2	`D1`
2	15	Order number
17	25	Item reference (subscription number)
42	5	Line number zero-padded (`00001`)
47	3	Spaces
50	18	Zeros
68	78	Spaces
146	28	Quantity block (`0000001` + 21 zeros)
174	12	Unit price in pence (always zero)
186	40	Spaces
226	13	ISBN / ISSN (13 digits)
239	27	Spaces

$$EOF — File Footer

Chars	Content
0–4	`$$EOF`
5–8	Sender code
9–10	Two spaces
11–17	File ID
18–20	Three spaces
21–34	Timestamp
35–41	Record count (7 digits, zero-padded) — H1 + H2 + H3 + D1 only

---

Output Filename Convention

```
{prefix}.{fileId}_{HHMM}_({DD-MM-YY}).txt
```

Example: `PO.0027816_1423_(24-02-26).txt`

XML Metadata Generator

How to Use

Step 1 — Download the template (first time only) Click **Download Metadata Template** in the Generate XML Metadata panel to save `metadata_template.xlsx`.

Step 2 — Populate the template Fill in one row per title. The template contains three columns: `ISSN`, `Title`, and `Page Extent`. All other specification values are calculated automatically.

Step 3 — Upload the file Drag the populated file onto the metadata upload zone or click to browse. The app validates the column structure and checks every ISSN value. If either check fails, the file is rejected with a detailed error report.

Step 4 — Generate Click **Generate XML & Download ZIP**. Two files are downloaded:

- `metadata.zip` — contains one `.xml` file per row, each named by its ISSN
- `metadata_summary.txt` — a plain-text summary report of all generated records

Step 5 — Clear (optional) Click **Clear** to reset the metadata panel and upload a new file.

Metadata Template — Column Reference

The metadata template contains exactly three columns. Column names and order must not be changed.

#	Column Header	Description

#	Column Header	Description
1	`ISSN`	13-digit ISSN with no spaces or hyphens. Used as the XML filename (`{ISSN}.xml`).
2	`Title`	Full journal or issue title. Written directly to `<title>` in the XML.
3	`Page Extent`	Total number of pages. Drives the paper type selection and spine size calculation.

ISSN Validation

After column structure validation passes, every ISSN value in the file is checked individually. An ISSN is valid only if it:

- Contains exactly **13 digits**
- Contains **no spaces**
- Contains **no hyphens**
- Contains **no other non-numeric characters**

If any ISSNs fail this check, the entire file is rejected and a row-by-row error report is shown listing the row number and the invalid value. The file must be corrected before generation can proceed.

Derived Values

Only ISSN, Title and Page Extent are read from the spreadsheet. All remaining specification values are fixed constants or calculated from Page Extent:

XML Field	Source	Value / Rule
`<trim_height>`	Fixed	`245`
`<trim_width>`	Fixed	`170`
`<colour>`	Fixed	`Mono`
`<binding_style>`	Fixed	`Limp`
`<lamination>`	Fixed	`Matt`
`<paper>`	Derived from Page Extent	`Magno Matt 130 gsm` if extent ≤ 38; `Magno Matt 90 gsm` if extent ≥ 39
`<gsm>`	Derived from Page Extent	`130` if extent ≤ 38; `90` if extent ≥ 39
`<spine>`	Calculated	See formula below
`<extent>`	From spreadsheet	Passed through directly

Spine Size Calculation

The spine size is calculated using the standard book spine formula, with a Limp binding addition applied:

```
spine = round( (pageExtent × gsm × volume) / 20000 + 0.65 )
```

Where:

- `gsm` is the paper grammage: `130` (for extent ≤ 38) or `90` (for extent ≥ 39)
- `volume` is `10` for both paper types
- `0.65` is the standard addition for Limp binding
- The result is rounded to the **nearest whole number** (millimetres)

Examples:

Page Extent	Paper Selected	Spine Calculation	Spine Size
16	Magno Matt 130 gsm	(16 × 130 × 10) / 20000 + 0.65	2 mm
32	Magno Matt 130 gsm	(32 × 130 × 10) / 20000 + 0.65	3 mm
39	Magno Matt 90 gsm	(39 × 90 × 10) / 20000 + 0.65	2 mm
120	Magno Matt 90 gsm	(120 × 90 × 10) / 20000 + 0.65	6 mm

XML Output Structure

Each generated XML file follows this structure:

```xml
<?xml version="1.0" encoding="UTF-8"?>
<record>
  <isbn>9771472645051</isbn>
  <title>[2026] 1 FCR 4</title>
  <trim_height>245</trim_height>
  <trim_width>170</trim_width>
  <extent>120</extent>
  <spine>6</spine>
  <paper>Magno Matt 90 gsm</paper>
  <gsm>90</gsm>
  <colour>Mono</colour>
  <binding_style>Limp</binding_style>
  <lamination>Matt</lamination>
</record>
```

Files are named `{ISSN}.xml` and bundled into `metadata.zip`.

Summary Report

A `metadata_summary.txt` file is downloaded separately alongside the ZIP. It contains a dated header, one entry per generated XML file, and a footer with totals. Example:

```
═══════════════════════════════════════════════════════════════════════
  BLOOMSBURY PUBLISHING — XML METADATA GENERATION SUMMARY
═══════════════════════════════════════════════════════════════════════
  Generated : 28 February 2026 at 09:45:12
  Source    : 3 rows processed
───────────────────────────────────────────────────────────────────────

    1. ISSN        : 9771472645051
       Title       : [2026] 1 FCR 4
       Page Extent : 120
       Paper Type  : Magno Matt 90 gsm
       Spine Size  : 6 mm
       Trim Size   : 245 × 170 mm
       Binding     : Limp  |  Lamination: Matt
       File        : 9771472645051.xml

───────────────────────────────────────────────────────────────────────
  Total XML files generated : 3
═══════════════════════════════════════════════════════════════════════
```

Column Validation

Both upload zones enforce strict column validation before any data is processed. The validation checks:

1. **Column count** — the file must have exactly the expected number of columns
2. **Column names** — each header must match the template exactly, including case, spacing, and punctuation
3. **Column order** — columns must appear in the same sequence as the template

If any check fails, the file is rejected immediately and a structured error panel is shown. Each mismatch is listed individually with its column number, the expected header, and what was actually found. No data is loaded until all errors are resolved.

For order uploads, a further **country code validation** step runs at generation time (see Country Code Validation).

For metadata uploads, a further **ISSN validation** step runs after the column check passes (see ISSN Validation).

The **Download Order Template** and **Download Metadata Template** buttons in each panel provide the correct template files to use as a starting point.

Template Downloads

All template and reference files must be present in the application's root directory (alongside `index.html`) for the download buttons to work.

Button	File	Panel

Button	File	Panel
Download Order Template	`order_file.xlsx`	Import Excel Data
Download User Guide	`USERGUIDE.pdf`	Import Excel Data
Download Metadata Template	`metadata_template.xlsx`	Generate XML Metadata

The template header rows must not be modified. The `USERGUIDE.pdf` is the converted version of `USERGUIDE.md` and should be regenerated from that source whenever the guide is updated.

Dependencies

All libraries are loaded from CDN and require an internet connection on first load. The browser will cache them for subsequent offline use.

Library	Version	Purpose	Source
SheetJS (xlsx)	0.18.5	Excel and CSV parsing	cdnjs.cloudflare.com
JSZip	3.10.1	ZIP file creation for XML bundle	cdnjs.cloudflare.com
Font Awesome	6.5.1	UI icons	cdnjs.cloudflare.com
IBM Plex Sans + Mono	—	Interface and monospace typography	fonts.googleapis.com

---

Browser Compatibility

Browser	Support
Chrome 90+	✅ Full support
Edge 90+	✅ Full support
Firefox 88+	✅ Full support
Safari 14+	✅ Full support
Internet Explorer	❌ Not supported

The Clipboard API requires HTTPS or `localhost`. On plain HTTP, copy-to-clipboard will silently fail — use the Download button instead.

Development Notes

EDI Field Positions

All character positions in `script.js` were verified by byte-level comparison against live production files:

- `T1.M02221600__22-02-26_.txt` (TFUK reference)
- `PO_TJ_20260223-190749-830.txt` (CUP reference)

Where the `hachette_order_file_format_spec.md` specification document disagreed with the actual files, the implementation follows the **actual files**.

Adding a New EDI Column

1. Add the exact header string to `ORDER_TEMPLATE_COLUMNS` in `script.js` at the correct position.
2. Add an entry to `EDI_FIELDS` with a unique `key`, display `label`, and `default` header name.
3. Add a `getCell(row, 'yourKey')` call at the appropriate character position inside `generateEDI()`.
4. Add the corresponding column to `order_file.xlsx` in the correct position, formatted as Text (`@`).
5. Update this README and `USERGUIDE.md`.

Changing Fixed XML Metadata Values

The fixed values (trim height, trim width, colour, binding style, lamination) are defined as constants inside `buildXML()` in `script.js`. Locate the `Fixed values` comment block and update the relevant string.

Changing the Paper / Extent Threshold

The extent threshold (currently ≤ 38 = Magno Matt 130 gsm, ≥ 39 = Magno Matt 90 gsm) is controlled by a single `if (extent <= 38)` condition inside `buildXML()`. The paper names, grammage values and volume constant (`10`) are all defined in the same function and can be updated independently.

Changing the Spine Formula

The spine calculation uses the formula `(extent × gsm × volume) / SPINE_FACTOR + LIMP_ADDITION` with `Math.round()`. The constants `SPINE_FACTOR` (20000), `LIMP_ADDITION` (0.65) and `VOLUME` (10) are declared as local constants inside `buildXML()` and can be adjusted there.

Changing the Carrier Code

The Royal Mail carrier code at H1 positions `[92:100]` is hardcoded as `'RMA  '` (8 characters). To change it, locate this line in `generateEDI()` and update the string, ensuring it remains exactly 8 characters:

```
+ 'RMA     '   // [92:100] Royal Mail carrier code
```

Updating the Supported Country Code List

The country list is defined in two places that must be kept in sync:

1. **`script.js`** — the `ISO_2_TO_3` object maps every supported 2-letter code to its alpha-3 equivalent. `VALID_ALPHA3` is derived automatically from its values. Add or remove entries here to change which codes are accepted at generation time.

2. **`order_file.xlsx`** — the Country Codes sheet contains the human-readable reference table, and column D of that sheet feeds the dropdown data validation on column M of the Order Form. Update both the table (columns A–C) and the validation list (column D) to match any changes made to `ISO_2_TO_3`.